

Application of Information & Communication Technologies

Lecture-7

Recap of Lecture 6

- ◆ Buses
- ◆ Memory
- ◆ ROM
- ◆ RAM
- ◆ Flash

Overview of Lecture 7

- ◆ Storage System Characteristics
- ◆ Primary Storage
- ◆ Secondary Storage
 - HDD & SSD

Characteristics of Storage

- ◆ What is Storage?
- ◆ Storage = saving data for future use
- ◆ RAM is temporary
- ◆ Storage is permanent
- ◆ Example:
 - RAM = whiteboard (erased easily).
 - Storage = notebook (keeps data until you delete).

Storage medium vs Storage device

◆ Medium

- Hardware where data is stored
- DVD disc, flash memory card, etc.

◆ Device

- DVD drive, flash memory card reader, etc.
- Medium is inserted into device to be used

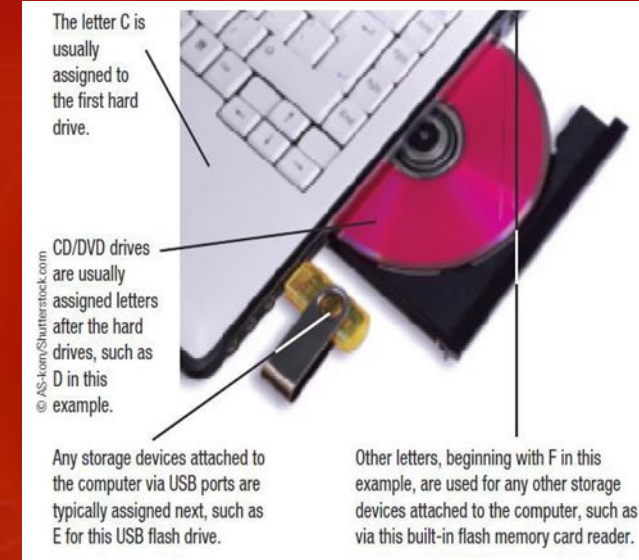


FIGURE 3-1
Storage device identifiers. To keep track of storage devices in an unambiguous way, the computer system assigns letters of the alphabet or names to each of them.

Storage vs Memory

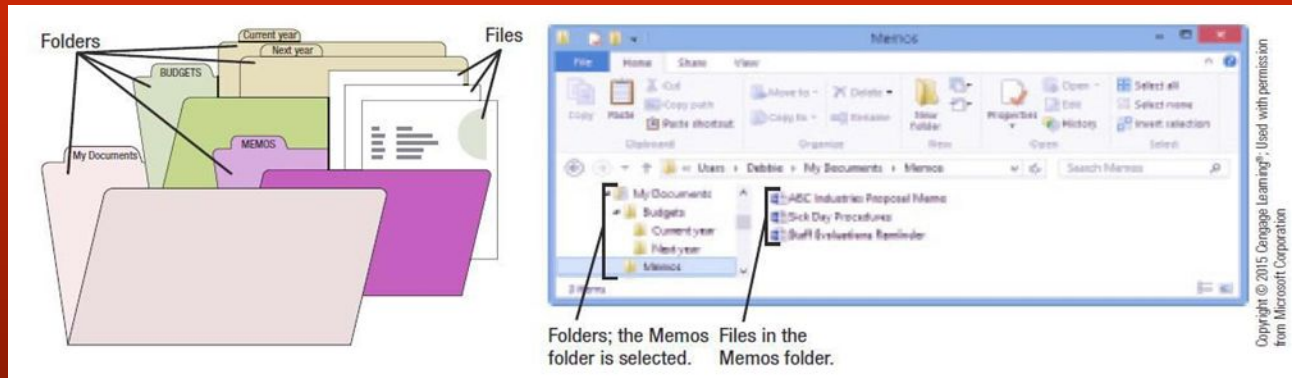
- ◆ Memory (RAM)
 - Temporary, fast, small
 - Loses data when power is off
 - Volatile
 - Example: Working desk (you keep things while working, but clear after)
- ◆ Storage (**HDD/SSD/USB**)
 - Permanent, slower, large
 - Keeps data even when power is off
 - Non-volatile
 - Example: Filing cabinet (data stays safe until removed)

Access Methods

- ◆ Random Access → data can be read from any place (HDD, SSD). Netflix / YouTube
- ◆ Sequential Access → data read in order (tape storage). Old cassette tapes / VHS
- ◆ Any Real Example ?
- ◆ Which access method is faster?

Logical vs Physical Storage

- ◆ Logical view → we see “files, folders, drives”.
- ◆ Physical view → computer sees “bits stored on disk”.



Storage Characteristics Summary

- ◆ Capacity → how much data (GB, TB).
- ◆ Speed → read/write time.
- ◆ Cost → price per GB.
- ◆ Durability → how long data lasts.

Primary Storage (Quick Recap)

- ◆ What is Primary Storage?
 - Directly used by CPU.
 - Includes RAM, Cache, ROM.

Features of Primary Storage

- ◆ Very fast.
- ◆ Small in size.
- ◆ Expensive.
- ◆ Volatile (RAM, Cache).

Secondary Storage

- ◆ What is Secondary Storage?
 - Used for long-term storage.
 - Examples: HDD, SSD, Optical discs, Flash drives.

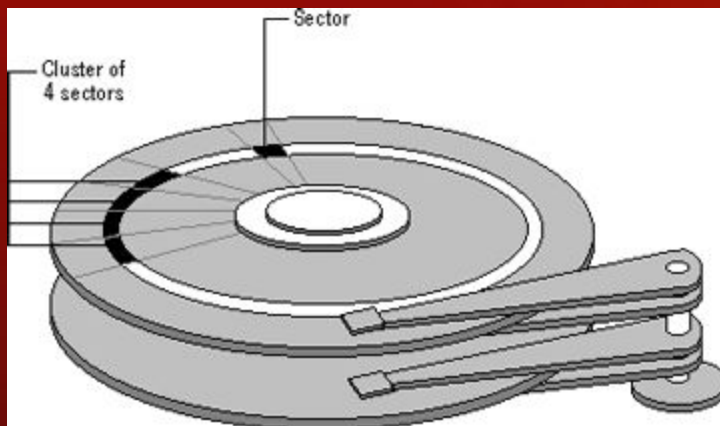
Hard Disk Drive (HDD)

- ◆ Magnetic storage.
- ◆ Large capacity, cheaper, slower.



Hard Disk Drive (HDD)

- ◆ Data stored in tracks, sectors, clusters.
- ◆ One or more permanently sealed metal magnetic disks with an access mechanism and read/write heads



Hard Disk Drive (HDD)

- ◆ **Hard disks are divided into:**
- ◆ **Tracks**
 - Concentric path on disk where data is recorded
- ◆ **Sectors**
 - A small piece of the track
- ◆ **Clusters**
 - One or more sectors; smallest addressable area of a disk
- ◆ **Cylinders**
 - Collection of tracks located in the same location on a set of hard disk surfaces

SECTORS

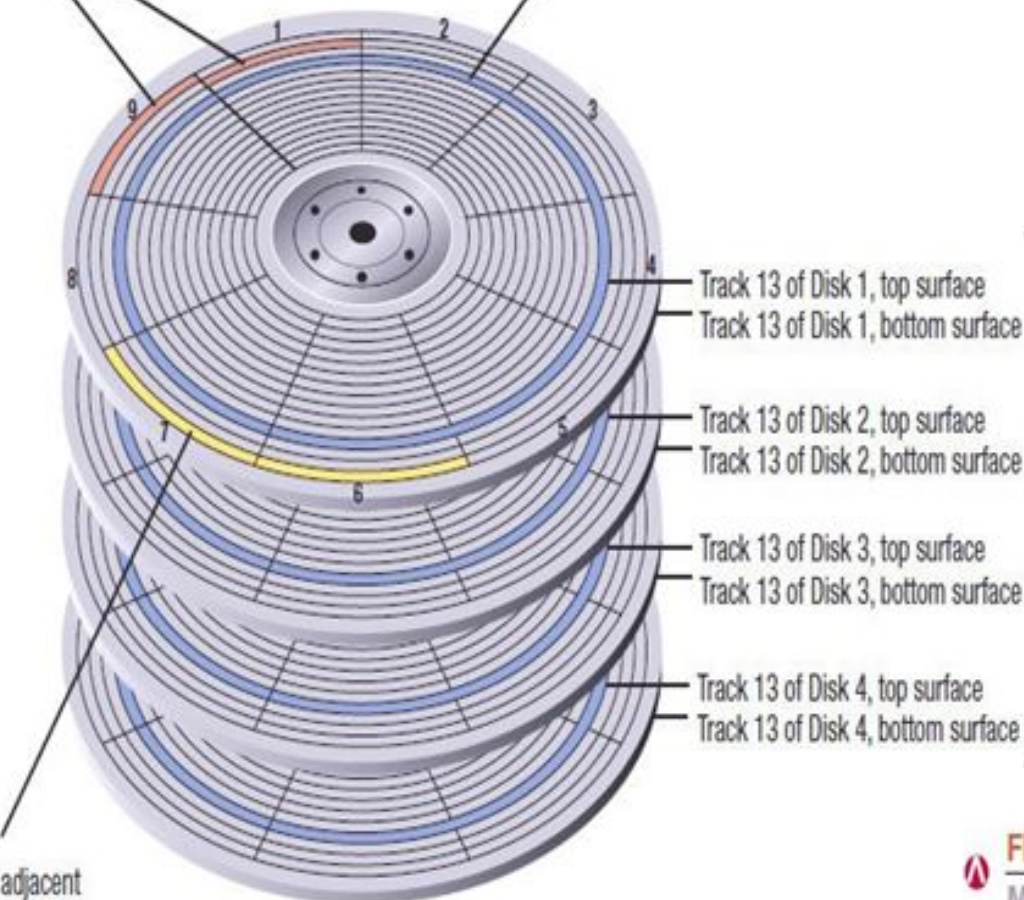
Each track is divided into sectors.

TRACKS

Data is stored on circular tracks; the 0s and 1s are represented magnetically.

CLUSTERS

One or more adjacent sectors form a cluster, the smallest amount of disk space that can be used to hold a file.



CYLINDER

A cylinder consists of a vertical stack of tracks, the same relative track on each disk surface.



FIGURE 3-6

Magnetic hard disks are organized into sectors, tracks, clusters, and cylinders.

Solid State Drives (SSDs)

- ◆ Use flash memory technology
- ◆ Use less power and have no moving parts
- ◆ Particularly appropriate for portable computers and mobile devices



Internal and External Hard Drives

- ◆ Internal hard drives : HDD or SSD located inside system unit
- ◆ External hard drives
 - Commonly used to transport large amounts of data from one computer to another
 - Portable external hard drives are smaller and easier to transport
 - Most connect with a USB connection although some may be wireless



FIGURE 3-8
External hard drives.

Summary

- ◆ Storage System Characteristics
- ◆ Primary Storage
- ◆ Secondary Storage
 - HDD & SDD

Suggested Reading

- ◆ Ch-03, The System Unit: Processing and Memory ,
“Understanding Computers: Today and Tomorrow,
Comprehensive”, 15th Edition by Deborah Morley
& Charles S. Parker
- ◆ Ch-06, Discovering Computers Fundamentals-
Your Interactive Guide -- Gary B Shelly; Misty E
Vermaat; Jeffrey J Q